

Smart Demand Signals

Inibsa · Interhack BCN 2026

1. Context and Motivation

Inibsa is a locally-based pharmaceutical manufacturing company, committed to European standards and sustainability criteria such as efficient use of water and energy, among others. This challenge seeks to generate a business intelligence solution to promote the sale of its locally produced products, adhering to these European standards and sustainability criteria. It is presented at the Interhack BCN Hackathon within the BCN Clima framework.

Inibsa operates with an approximate base of 7,000 dental clinics and has a sales history spanning more than five years at the client, product, and date level. This context allows tackling a high-value commercial challenge: transforming purchasing patterns into actionable predictive signals.

The case presents two clearly different dynamics. On one hand, there are commoditised product categories with recurring purchases, tied to the clinic's habitual consumption — such as anaesthesia, needles, and disinfection products. In these families, not all clients show the same degree of engagement: some buy marginally or residually, others concentrate most of their demand with the company, and others alternate their purchases with competitors. The challenge is to detect these differences and, above all, identify the optimal moment of contact to capture demand that is not currently being materialised.

On the other hand, there are technical products whose purchase depends more heavily on the type of clinical case, the professional's specialty, and competitor penetration within the account. In this area, the priority is not only to anticipate replenishment, but to detect early signals of deterioration in purchasing patterns or risk of abandonment.

2. Challenge Objective

Design an analytical solution capable of identifying, on a daily basis, signals indicating the need for commercial intervention at the level of client, product family, and timing — differentiating between two main use cases: commodity products and technical products.

The company is looking for a solution capable of converting transactional and commercial data into interpretable and actionable alerts, directed to the appropriate channel in each case: sales representative, telesales, or marketing automation. The solution should initially be designed as standalone, with the possibility of future integration into a CRM or other platforms, without depending on a specific technology.

In this way, the solution contributes to promoting the sale of Inibsa's locally produced products, following European standards and sustainability criteria such as efficient use of water and energy, while better anticipation of demand implies a more efficient supply chain.

3. Technical Challenge / Functional Requirements

The solution must treat commodity products with recurring purchases and technical products with more variable patterns differently. For commodities, it must estimate each client's expected purchasing behaviour relative to their consumption potential, distinguishing between clients with zero or marginal purchases, loyal clients, and promiscuous clients who split their demand with competitors. For technical products, it must identify signals of risk of loss or abandonment, detecting drops in frequency, drops in volume, total disappearance of purchases, or anomalous activity relative to the client's historical pattern.

The system must recalculate signals and alerts on a daily basis, on a stable and maintainable update foundation. Each alert must include a contact recommendation, reason, and affected product family, and must offer the ability to reconstruct why it was generated and which variables or rules were involved in its activation.

The solution should not be limited to generating alerts, but must help to order and operationalise them, incorporating expected economic impact, conversion probability, and time urgency. The resulting action may be directed to a sales representative, telesales, or marketing platform (e.g. HubSpot), depending on the commercial assignment of the client. The proposal must be agnostic with respect to specific CRMs and must contemplate what happens after the alert: who manages it, within what timeframe, and how its result is recorded.

4. Available Resources and Data

Inibsa has information at the client and product level with daily granularity, purchase channel, representative or telesales activity, client data (purchasing potential, location, client type), and some indirect and qualitative indications about competitors.

The solution must explicitly address aspects of data quality, consistency, and traceability — managing clients with incomplete histories, normalisation of product families, product changes or substitutions, and clients with irregular or seasonal purchases. It must also handle anomalies that could distort the signal, such as extraordinary orders, promotions, stock breaks, or changes in commercial policy. Part of the CRM commercial activity is in free-text format, so its use should be considered optional or exploratory. Data reliability is higher in Spain than in Portugal, which is why the initial development will focus on Spain.

5. Expected Deliverables

Each team must develop a proposal capable of translating the available data into a useful tool for commercial action. Ideally, the solution will offer: prediction of

purchase need for commodities, early detection of churn risk for technical products, clear identification of capture windows against competitors, interpretable and actionable alerts, operational prioritisation, a proposal for daily operation, and an architecture to operate standalone and later evolve towards CRM integration.

The solution must also contemplate the capacity to learn from its own use, by recording the generated alert, the commercial action taken, and the result obtained, so as to improve the model, detect false positives, and adjust rules or parameters over time.

6. Evaluation Criteria

Proposals will be especially valued on: the ability to translate a complex business problem into a useful analytical solution; intelligent differentiation between commodity and technical products; quality of the predictive logic; interpretability and traceability of alerts; prioritisation capability and fit within the commercial process; real applicability in a commercial environment; and scalability and viability of future evolution.

Damm Smart Truck

Damm · Interhack BCN 2026

1. Context and Motivation

Damm has a highly complex logistics operation linked to the distribution of products to multiple channels and points of sale. In this context, DDI (Direct Integral Distribution) plays a key role as the group's distribution network, with a multi-product catalogue and service to hospitality industry clients and other channels.

Each truck can deliver to approximately 15 to 25 clients on a route. Currently, the load is prepared mainly by grouping products by reference, as this facilitates warehouse work and optimises truck space. However, during delivery, the driver may need to search for each client's products at different points in the vehicle, which can generate additional movements and time losses.

Furthermore, the delivery order is not fully predefined: it is often the driver who decides the route. This opens an opportunity to apply data, optimisation, and artificial intelligence — combining two dimensions that are usually treated separately: the delivery route and the physical preparation of the truck's load.

The challenge must also take into account the operational reality of DDI: different product formats and volumes, limited capacity, lateral access to pallets via side curtains, handling restrictions, and reverse logistics. Approximately 60% of the products delivered are returnable, meaning trucks also collect empty crates, containers, or barrels during the route.

2. Challenge Objective

The objective is to design a solution that allows jointly optimising the delivery route and the truck's load configuration, taking into account the delivery order, volume and type of products, vehicle capacity, reverse logistics, and operational restrictions.

The ideal solution should propose an efficient route and recommend how to physically prepare the load so that unloading is more agile, without excessively penalising space utilisation or warehouse preparation. The goal is not just to find the shortest route, but the best balance between route efficiency, load efficiency, ease of unloading, and operational viability.

3. Technical Challenge / Functional Requirements

The solution must combine route optimisation with load optimisation. It must propose the order of client visits based on variables such as distance, estimated time, number of clients, time windows, or priorities — and simultaneously recommend how to physically distribute products within the truck according to volume, type, capacity, and planned unloading order.

It will also be necessary to explore which loading model is most efficient: the current grouping by references, a client-oriented load, or a hybrid model. It will be valued if the proposal includes a simplified visualisation of the load (zones, pallets, product groups, associated clients, or unloading order) and incorporates reverse logistics.

The solution must respect real operational restrictions, such as lateral access to pallets via side curtains, volume or weight limitations, safe handling, and the viability of the warehouse preparation process. It will also be valued if it generates explainable recommendations: optimal loading order, clients that are convenient to group, critical products, or expected friction points.

4. Available Resources and Data

Damm and DDI will provide contextual information and, where possible, sample or synthetic data: examples of orders, clients, references, quantities, formats, volumes, or approximate logistical units.

Examples of routes or client-to-driver/truck assignments may also be shared, along with information about how DDI currently operates, visual materials of the warehouse and current product distribution inside trucks, and simplified operational restrictions on capacity, lateral access, volume, weight, handling, and safety.

Information about reverse logistics will be included, such as the approximate percentage of returnable products and examples of typical returns. If real data cannot be shared, anonymised, simplified, or synthetic data will be provided. During the event, Damm and DDI will aim to have Distribution and Data & AI personnel available to resolve operational and technical queries.

5. Expected Deliverables

Each team must present a proposal that allows understanding both the technical solution and its practical application in the Damm and DDI context.

A functional prototype or conceptual demo (tool, model, simulator, script, dashboard, or interface) is expected, showing how route and load could be jointly optimised. It will also be necessary to explain the decision logic used: optimisation criteria, variables considered, model restrictions, and priorities.

The proposal must include a visualisation of the route and the load, applied to a specific use case with a simulated or sample route. The final pitch must summarise the problem, the solution, the assumptions, the limitations, the results, and the possible next steps towards a real pilot.

6. Evaluation Criteria

Proposals will be evaluated combining technical, operational, and communication criteria. Special importance will be given to: real applicability in the Damm and DDI context, technical quality of the optimisation approach, potential impact on efficiency, time, sustainability, service quality or ergonomics, creativity of the proposal, and clarity of the pitch.

Indicative weightings: 30% real applicability, 25% technical quality, 20% potential impact, 15% creativity, and 10% clarity of communication.

EdgeAI for a Greener Barcelona

Qualcomm · Interhack BCN 2026

1. Context and Motivation

Barcelona is undergoing a profound transformation. The mass of tourists visiting the city coincides simultaneously with the realities of climate change and the needs of its citizens. Extreme heat waves, water scarcity, loss of urban green spaces, inefficient buildings, and growing mobility demands pose challenges that require more adaptive solutions.

Traditional approaches are no longer sufficient to keep Barcelona as a sustainable and resilient city in the face of these demands. This is where EdgeAI with Arduino and Edge Impulse, using Qualcomm technology, presents a unique opportunity: we can enable real-time monitoring, autonomous decision-making, and localised actions with minimal energy consumption and latency.

This challenge invites participants to use Arduino UNO Q, sensors, and Edge Impulse to build innovative on-device AI solutions that help Barcelona become a more climate-resilient, efficient, and future-ready city.

2. Challenge Objective

Build an EdgeAI system with Arduino UNO Q and Edge Impulse capable of: collecting and processing data from sensors, cameras, or existing hardware interfaces; running efficient local inference (classification, object detection, anomaly detection, or small multimodal models); making autonomous decisions and triggering local actions; and operating under real-world constraints such as limited power, memory, and intermittent connectivity.

3. Technical Challenge / Application Areas

The challenge can be approached within one or more of the following application areas, all aligned with Barcelona's urban challenges:

- **Revitalising urban nature:** Build tools to monitor, protect, and improve urban biodiversity, green corridors, and the city's broader natural systems.
- **Efficient buildings:** Design EdgeAI systems capable of measuring and improving the energy performance of existing buildings and supporting the transition to the future climate.
- **Energy transition and optimisation:** Facilitate the shift towards a cleaner energy future by developing EdgeAI solutions for local energy monitoring and consumption optimisation.
- **Future-ready mobility:** Develop EdgeAI solutions that improve urban mobility, reduce traffic-related emissions, and increase safety and efficiency for all users.

4. Available Resources and Hardware

Teams will work with real technology. The available hardware and tools are: Arduino UNO Q boards and Arduino App Lab, USB cameras, modulinos and environmental sensors, as well as a free Edge Impulse Studio account.

During the event, teams will have direct access to Edge Impulse experts and the Arduino team, who can resolve technical queries, help with hardware setup, and provide guidance on best practices for developing efficient models on resource-constrained devices.

5. Expected Deliverables

Each team must present a functional EdgeAI solution deployed on the provided hardware. The proposal must demonstrate how the system collects data, processes information locally, makes autonomous decisions, and triggers actions without constantly depending on an external connection.

The final pitch should: show the prototype operating in real time (demo with real sensors or data), explain the model logic and technical decisions made, and place the solution within one of the defined application areas — clearly showing its potential impact on one of Barcelona's climate or urban challenges.

6. Evaluation Criteria

Proposals will be evaluated especially on three main criteria: the creativity of the solution and its real relevance to Barcelona's urban challenges; technical execution and model efficiency, assessing aspects such as latency, energy consumption, and accuracy; and the potential impact and future scalability of the project beyond the prototype presented during the hackathon.